



Session #2-3: Multi-armed Bandits

Instructors - DRL Course

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Agenda for the class

- Recap
- k-armed Bandit Problem & its significance
- Action-Value Methods
 - Sample Average Method & Incremental Implementation
- Non-stationary Problem
- Initial Values & Action Selection



K-armed Bandit Problem

Problem

- You are faced repeatedly with a choice among k different options, or actions
- After each choice of actions you receive a numerical reward
Reward is chosen from a stationary probability distribution that depends on the selected action
- **Objective** : to *maximize the expected total reward over some time period*

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Strategy:

- Identify the best lever(s)
- Keep pulling the identified ones

Questions:

- How do we define the **best ones**?
- What are the best levers?

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$$\mathbb{E}[a] = -\$0.5$$



$$\mathbb{E}[b] = -\$0.2$$



$$\mathbb{E}[c] = \$0.1$$



$$\mathbb{E}[d] = \$0.11$$

- Expected Mean Reward** for each action selected
→ call it **Value** of the action

$$q_*(a) \doteq \mathbb{E}[R_t \mid A_t = a]$$

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K-armed Bandit Problem

$$q_*(a) \doteq \mathbb{E}[R_t \mid A_t = a]$$

- A_t - action selected on time step t
- $Q_t(a)$ - estimated value of action a at time step t
- $q_*(a)$ - value of an arbitrary action a

Note: If you knew the value of each action, then it would be trivial to solve the k -armed bandit problem: you would always select the action with highest value :-)

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K-armed Bandit Problem



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$$\mathbb{E}[b] = -\$0.2$$



$$\mathbb{E}[c] = \$0.1$$



$$\mathbb{E}[d] = \$0.11$$

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K-armed Bandit Problem



$$-1, -1, 5$$

$$\hat{\mathbb{E}}[a] = 1$$



$$-0.2, -0.2$$

$$\hat{\mathbb{E}}[b] = -0.2$$



$$-0.5, -0.5, -0.5$$

$$\hat{\mathbb{E}}[c] = -0.5$$



$$-2, -2$$

$$\hat{\mathbb{E}}[d] = -2$$

Keep pulling the levers; update the estimate of action values;

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K-armed Bandit Problem



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$$-2, -2$$

$$\hat{\mathbb{E}}[d] = -2$$

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K-armed Bandit Problem

1. How to maintain the estimate of expected rewards for each action?

Average the rewards actually received !!!

$$Q_t(a) \doteq \frac{\text{sum of rewards when } a \text{ taken prior to } t}{\text{number of times } a \text{ taken prior to } t}$$

$$= \frac{\sum_{i=1}^{t-1} R_i \cdot \mathbb{1}_{A_i=a}}{\sum_{i=1}^{t-1} \mathbb{1}_{A_i=a}}$$

1. How to use the estimate in selecting the right action?

Greedy Action Selection $A_t \doteq \underset{a}{\operatorname{argmax}} Q_t(a)$

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K-armed Bandit Problem

- How to use the estimate in selecting the right action?

Greedy Action Selection

$$A_t \doteq \arg \max_a Q_t(a)$$

Actions which are inferior by the value estimate upto time t, could be indeed better than the greedy action at t !!!

- Exploration vs. Exploitation?

ϵ -Greedy Action Selection / near-greedy action selection

Behave greedily most of the time; Once in a while, with small probability ϵ select randomly from among all the actions with equal probability, independently of the action-value estimates.

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K-armed Bandit Problem



$\{-1, -1, 5\}$
 $N_{11}(a)=3$
 $q_*(a)=-\$0.5$
 $Q_{11}(a)=1$



$\{-0.2, -0.2\}$
 $N_{11}(b)=2$
 $q_*(b)=-\$0.2$
 $Q_{11}(b)=-0.2$



$\{-0.5, -0.5, -0.5\}$
 $N_{11}(c)=3$
 $q_*(c)=$0.1$
 $Q_{11}(c)=-0.5$



$\{-2, -2\}$
 $N_{11}(d)=2$
 $q_*(d)=$1$
 $Q_{11}(d)=-2$

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K-armed Bandit Problem

Greedy Action



$$\begin{aligned} &\{-1, -1, 5\} \\ &N_{11}(a)=3 \\ &q_*(a)=-\$0.5 \\ &Q_{11}(a)=1 \end{aligned}$$



$$\begin{aligned} &\{-0.2, -0.2\} \\ &N_{11}(b)=2 \\ &q_*(b)=-\$0.2 \\ &Q_{11}(b)=-0.2 \end{aligned}$$



$$\begin{aligned} &\{-0.5, -0.5, -0.5\} \\ &N_{11}(c)=3 \\ &q_*(c)=$0.1 \\ &Q_{11}(c)=-0.5 \end{aligned}$$



$$\begin{aligned} &\{-2, -2\} \\ &N_{11}(d)=2 \\ &q_*(d)=$1 \\ &Q_{11}(d)=-2 \end{aligned}$$

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K-armed Bandit Problem

Action to Explore



$$\begin{aligned} &\{-1, -1, 5\} \\ &N_{11}(a)=3 \\ &q_*(a)=-\$0.5 \\ &Q_{11}(a)=1 \end{aligned}$$



$$\begin{aligned} &\{-0.2, -0.2\} \\ &N_{11}(b)=2 \\ &q_*(b)=-\$0.2 \\ &Q_{11}(b)=-0.2 \end{aligned}$$



$$\begin{aligned} &\{-0.5, -0.5, -0.5\} \\ &N_{11}(c)=3 \\ &q_*(c)=$0.1 \\ &Q_{11}(c)=-0.5 \end{aligned}$$



$$\begin{aligned} &\{-2, -2\} \\ &N_{11}(d)=2 \\ &q_*(d)=$1 \\ &Q_{11}(d)=-2 \end{aligned}$$

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K-armed Bandit Problem

ϵ -Greedy Action Selection / near-greedy action selection

```
epsilon = 0.05 // small value to control exploration
def get_action():
    if random.random() > epsilon:
        return argmaxa(Q(a))
    else:
        return random.choice(A)
```

- In the limit as the number of steps increases, every action will be sampled by ϵ -greedy action selection an infinite number of times. This ensures that all the $Q_t(a)$ converge to $q_*(a)$.
- Easy to implement / optimize for epsilon / yields good results

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Ex-1: In ϵ -greedy action selection, for the case of two actions and $\epsilon = 0.5$, what is the probability that *the greedy action* is selected?

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Ex-1: In ϵ -greedy action selection, for the case of two actions and $\epsilon = 0.5$, what is the probability that *the greedy action* is selected?

$p(\text{greedy action})$

$$\begin{aligned}
 &= p(\text{greedy action AND greedy selection}) + p(\text{greedy action AND random selection}) \\
 &= p(\text{greedy action} \mid \text{greedy selection}) p(\text{greedy selection}) \\
 &\quad + p(\text{greedy action} \mid \text{random selection}) p(\text{random selection}) \\
 &= p(\text{greedy action} \mid \text{greedy selection}) (1-\epsilon) + p(\text{greedy action} \mid \text{random selection}) (\epsilon) \\
 &= p(\text{greedy action} \mid \text{greedy selection}) (0.5) + p(\text{greedy action} \mid \text{random selection}) (0.5) \\
 &= (1) (0.5) + (0.5) (0.5) \\
 &= 0.5 + 0.25 \\
 &= 0.75
 \end{aligned}$$

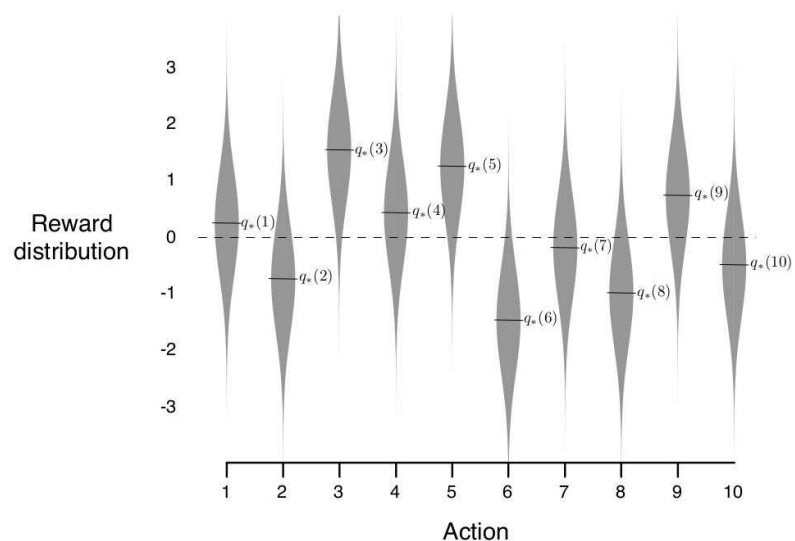
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10-armed Testbed

Example:

- A set of 2000 randomly generated k -armed bandit problems with $k = 10$
- Action values were selected according to a normal (Gaussian) distribution with mean 0 and variance 1.
- While selecting action A_t at time step t , the actual reward, R_t , was selected from a normal distribution with mean $q_*(A_t)$ and variance 1
- **One Run** : Apply a method for 1000 time steps to one of the bandit problems
- Perform 2000 runs, each run with a different bandit problem, to get an algorithms average behavior

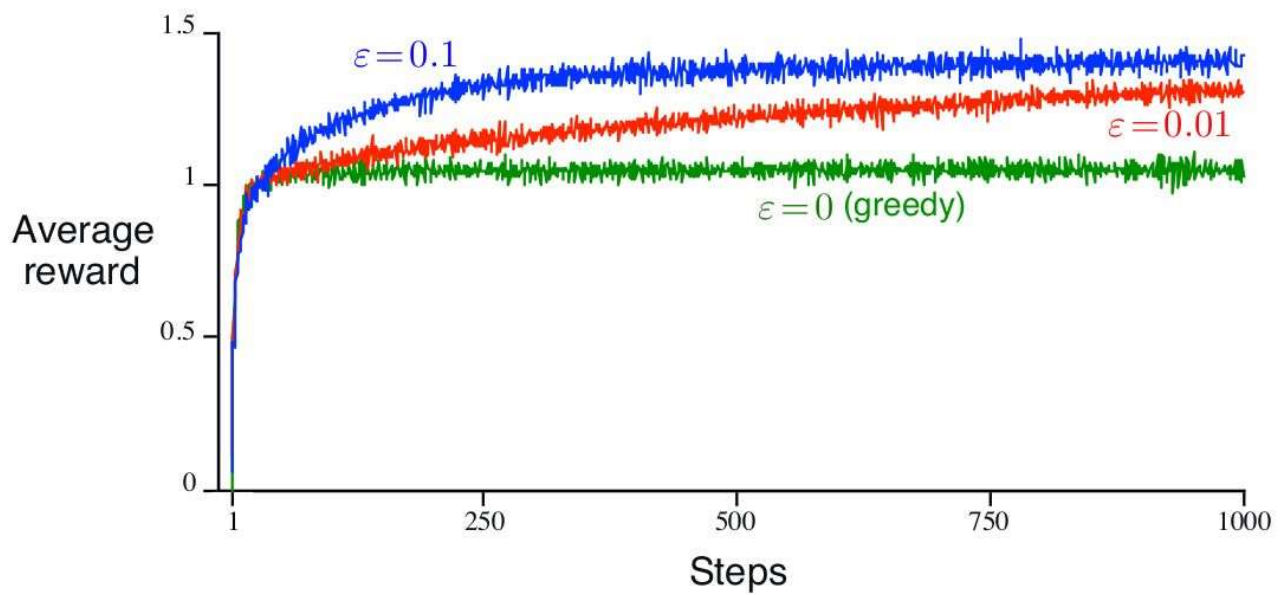


An example bandit problem from the 10-armed testbed

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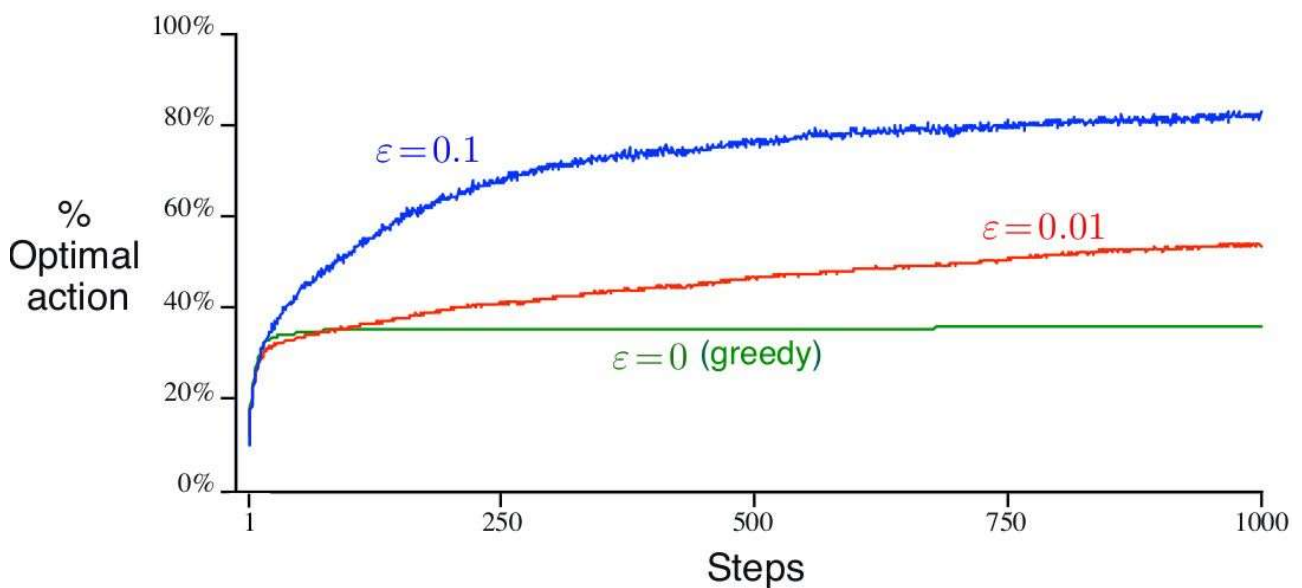
Average performance of ϵ -greedy action-value methods on the 10-armed testbed



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Average performance of ϵ -greedy action-value methods on the 10-armed testbed



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Discussion on Exploration vs. Exploitation

- 1) What if the reward variance is
 - a. larger, say 10 instead of 1?
 - b. zero ? [deterministic]
- 2) What if the bandit task is non-stationary? [that is, the true values of the actions changed over time]

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Ex-2:

Consider a k -armed bandit problem with $k = 4$ actions, denoted 1, 2, 3, and 4.

Consider applying to this problem a bandit algorithm using ϵ -greedy action selection, sample-average action-value estimates, and initial estimates of $Q_1(a) = 0$, for all a .

Suppose the initial sequence of actions and rewards is $A_1 = 1, R_1 = 1, A_2 = 2, R_2 = 1, A_3 = 2, R_3 = 2, A_4 = 2, R_4 = 2, A_5 = 3, R_5 = 0$.

On some of these time steps the ϵ case may have occurred, causing an action to be selected at random.

On which time steps did this definitely occur? On which time steps could this possibly have occurred?

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Incremental Implementation

- Efficient approach to compute the estimate of action-value;

$$Q_n \doteq \frac{R_1 + R_2 + \dots + R_{n-1}}{n-1}.$$

- Given Q_n and the n th reward, R_n , the new average of all n rewards can be computed as follows

$$\begin{aligned} Q_{n+1} &= \frac{1}{n} \sum_{i=1}^n R_i \\ &= \frac{1}{n} \left(R_n + \sum_{i=1}^{n-1} R_i \right) \\ &= \frac{1}{n} \left(R_n + (n-1) \frac{1}{n-1} \sum_{i=1}^{n-1} R_i \right) \\ &= \frac{1}{n} (R_n + (n-1)Q_n) \\ &= \frac{1}{n} (R_n + nQ_n - Q_n) \\ &= Q_n + \frac{1}{n} [R_n - Q_n], \end{aligned}$$

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Incremental Implementation

Note:

- StepSize decreases with each update
- We use α or $\alpha_t(a)$ to denote step size (constant / varies with each step)

Discussion:

Const vs. Variable step size?

$$\begin{aligned} Q_{n+1} &= \frac{1}{n} \sum_{i=1}^n R_i \\ &= \frac{1}{n} \left(R_n + \sum_{i=1}^{n-1} R_i \right) \\ &= \frac{1}{n} \left(R_n + (n-1) \frac{1}{n-1} \sum_{i=1}^{n-1} R_i \right) \\ &= \frac{1}{n} (R_n + (n-1)Q_n) \\ &= \frac{1}{n} (R_n + nQ_n - Q_n) \\ &= Q_n + \frac{1}{n} [R_n - Q_n], \end{aligned}$$

$$NewEstimate \leftarrow OldEstimate + StepSize [Target - OldEstimate]$$

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Bandit Algorithm with Incremental Update/ ϵ -greedy selection

Initialize, for $a = 1$ to k :

$$Q(a) \leftarrow 0$$

$$N(a) \leftarrow 0$$

Loop forever:

$$A \leftarrow \begin{cases} \arg \max_a Q(a) & \text{with probability } 1 - \epsilon \\ \text{a random action} & \text{with probability } \epsilon \end{cases} \quad (\text{breaking ties randomly})$$

$$R \leftarrow \text{bandit}(A)$$

$$N(A) \leftarrow N(A) + 1$$

$$Q(A) \leftarrow Q(A) + \frac{1}{N(A)} [R - Q(A)]$$

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What are problems whose solutions are modelled as MAB?

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Non-stationary Problem

- Most RL problems are non-stationary !
- Give more weight to recent rewards than to long-past rewards !!!

$$Q_{n+1} \doteq Q_n + \alpha [R_n - Q_n]$$

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Non-stationary Problem

- Most RL problems are non-stationary !
- Give more weight to recent rewards than to long-past rewards !!!

$$Q_{n+1} \doteq Q_n + \alpha [R_n - Q_n]$$

Exponential recency-weighted average

$$\begin{aligned} Q_{n+1} &= Q_n + \alpha [R_n - Q_n] \\ &= \alpha R_n + (1 - \alpha) Q_n \\ &= \alpha R_n + (1 - \alpha) [\alpha R_{n-1} + (1 - \alpha) Q_{n-1}] \\ &= \alpha R_n + (1 - \alpha) \alpha R_{n-1} + (1 - \alpha)^2 Q_{n-1} \\ &= \alpha R_n + (1 - \alpha) \alpha R_{n-1} + (1 - \alpha)^2 \alpha R_{n-2} + \\ &\quad \dots + (1 - \alpha)^{n-1} \alpha R_1 + (1 - \alpha)^n Q_1 \\ &= (1 - \alpha)^n Q_1 + \sum_{i=1}^n \alpha (1 - \alpha)^{n-i} R_i. \end{aligned}$$

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Optimistic Initial Values

- All the above discussed methods are **biased** by their initial estimates
- For sample average method the bias disappears once all actions have been selected at least once
- For methods with constant α , the bias is permanent, though decreasing over time
- Initial action values can also be used as a simple way of encouraging exploration.
- In 10 armed testbed, set initial estimate to +5 rather than 0.

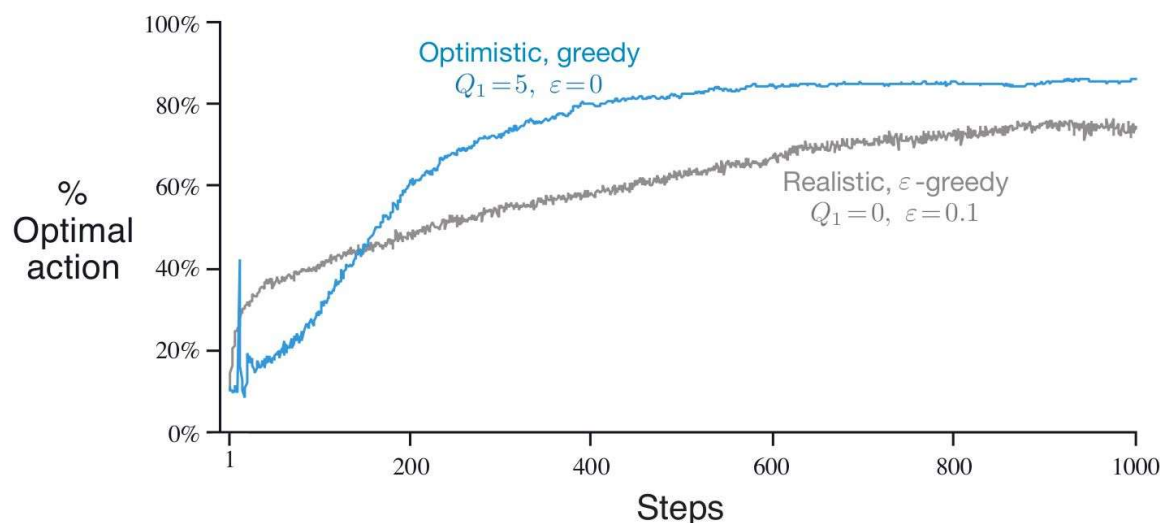
This can encourage action-value methods to explore.

Whichever actions are initially selected, the reward is less than the starting estimates; the learner switches to other actions, being disappointed with the rewards it is receiving. The result is that all actions are tried several times before the value estimates converge.

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Optimistic Initial Values



Caution:

Optimistic Initial Values can only be considered as a simple trick that can be quite effective on stationary problems, but it is far from being a generally useful approach to encouraging exploration.

Question:

Explain how in the non-stationary scenario the optimistic initial values will fail (to explore adequately).

The effect of optimistic initial action-value estimates on the 10-armed testbed. Both methods used a constant step-size parameter, $\alpha = 0.1$

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Upper-Confidence-Bound Action Selection

- ϵ -greedy action selection forces the non-greedy actions to be tried, Indiscriminately, with no preference for those that are nearly greedy or particularly uncertain
- It would be better to select among the non-greedy actions according to their potential for actually being optimal
Take into account both how close their estimates are to being maximal and the uncertainties in those estimates.

$$A_t \doteq \arg \max_a \left[Q_t(a) + c \sqrt{\frac{\ln t}{N_t(a)}} \right]$$

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Upper-Confidence-Bound Action Selection

- Each time a is selected the uncertainty is presumably reduced
- Each time an action other than a is selected, t increases but $N_t(a)$ does not; because t appears in the numerator, the uncertainty estimate increases.
- Actions with lower value estimates, or that have already been selected frequently, will be selected with decreasing frequency over time

Diagram illustrating the components of the Upper-Confidence-Bound (UCB) action selection formula:

$$A_t \doteq \arg \max_a \left[Q_t(a) + c \sqrt{\frac{\ln t}{N_t(a)}} \right]$$

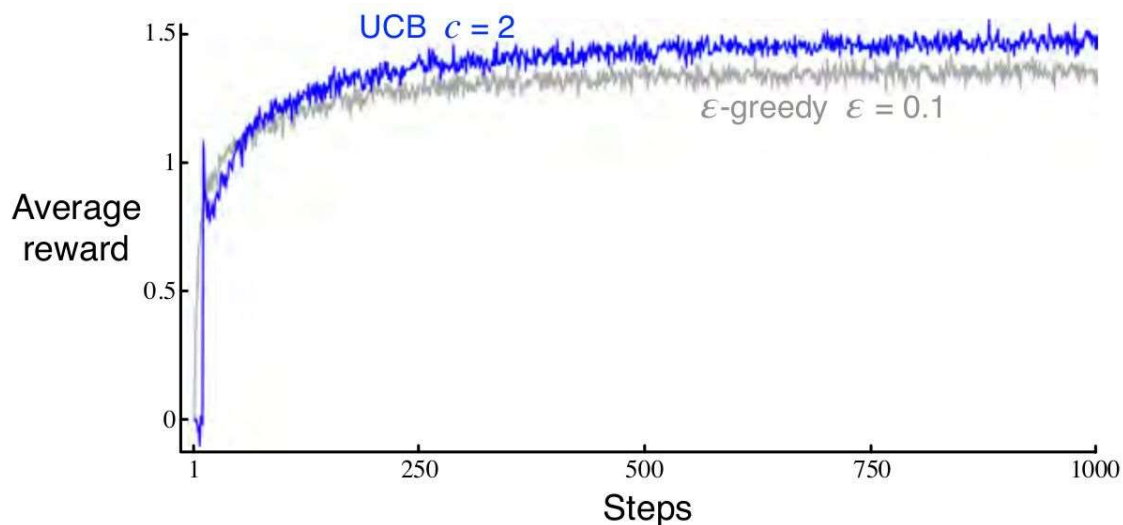
The diagram shows three labels with arrows pointing to parts of the formula:

- Action Value at time t for a** points to $Q_t(a)$.
- Confidence Level** points to the constant c .
- Measure of Uncertainty** points to the square root term $\sqrt{\frac{\ln t}{N_t(a)}}$.

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Upper-Confidence-Bound Action Selection



UCB often performs well, as shown here, but is more difficult than ϵ -greedy to extend beyond bandits to the more general reinforcement learning settings



Policy-based algorithms

- Forget about action-value (**Q**) estimates, we don't really care about them
- We care about what actions to choose
 - Let's assign a preference to each action and tweak its value
- Define $H_t(a)$ as a numerical preference value associated with action a
- Which action is selected?
 - $A_t = \underset{a}{\operatorname{argmax}} [H_t(a)]$
 - Hardmax results in no exploration -- deterministic action selection



Softmax!



Softmax function

- **Input:** vector of preferences
- **Output:** vector of probabilities forming a valid distribution
- $\Pr\{a_t = a\} = \frac{e^{H_t(a)}}{\sum_{a' \in A} e^{H_t(a')}} = \Pr(a)$
- $H_t \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{bmatrix} 6 \\ 9 \\ 2 \end{bmatrix}$
- $\text{softmax} \begin{pmatrix} 6 \\ 9 \\ 2 \end{pmatrix} = \begin{bmatrix} 0.047 \\ 0.952 \\ 0.00087 \end{bmatrix}$
- That is, with $\Pr(0.95)$ choose a_2 , $\Pr(0.05)$ choose a_1 , and $< \Pr(0.01)$ choose a_3

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Softmax function

- Exploration – checked!
- Softmax provides another important attribute – **a differentiable policy**
- Say that we learn that action a_1 results in good relative performance
- Hardmax: $A_t = \underset{a}{\operatorname{argmax}}[H_t(a_1), H_t(a_2), H_t(a_3)]$
 - Change $H(a_1)$ such that $\Pr(a_1)$ is increased, $\frac{\partial \Pr(a_1)}{\partial H(a_1)} = NA$
- Softmax: $\Pr(a_1) = \frac{e^{H_t(a_1)}}{\sum_{a' \in A} e^{H_t(a')}}$
 - Change $H(a_1)$ such that $\Pr(a_1)$ is increased $\rightarrow$ update towards $\frac{\partial \Pr(a_1)}{\partial H(a_1)}$

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Gradient ascend

- $H_{t+1}(A_t) = H_t(A_t) + \alpha(R_t - \bar{R}_t) \frac{\partial \Pr(A_t)}{\partial H(A_t)}$?
- $\forall a \neq A_t, H_{t+1}(a) = H_t(a) + \alpha(R_t - \bar{R}_t) \frac{\partial \Pr(A_t)}{\partial H(a)}$
- Update the preferences based on observed reward and a baseline reward ($\bar{R}_t$)
- If the observed reward is larger than the baseline:
 - Increase the preference of the chosen action, A_t
 - Decrease the preference of all other actions, $\forall a \neq A_t$
- Else do the opposite

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Update Rule

On each step, after selecting action A_t and receiving the reward R_t ,
Update the action preferences :

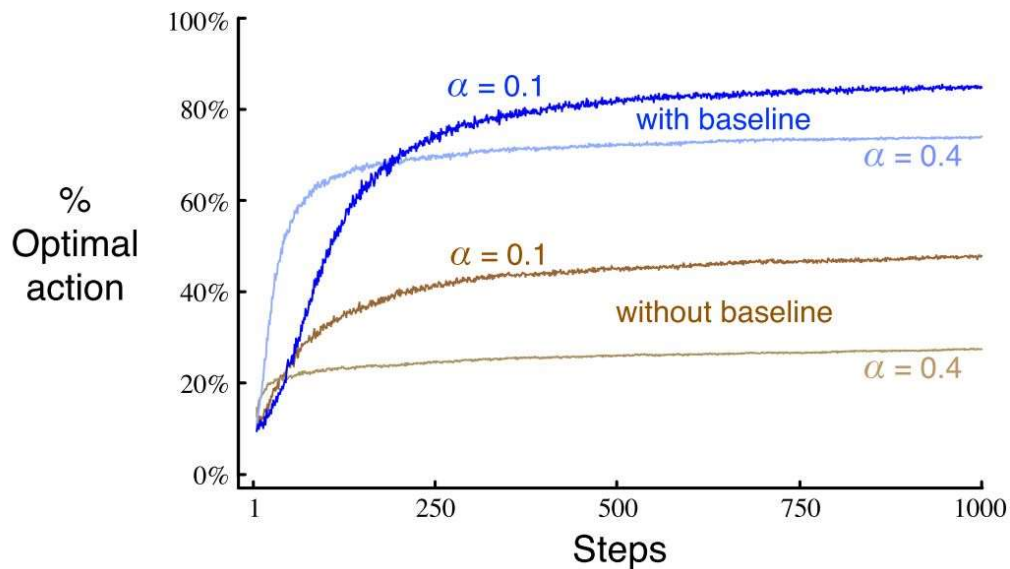
$$H_{t+1}(A_t) = H_t(A_t) + \alpha(R_t - \bar{R}_t)(1 - \Pr(A_t))$$

$$\forall a \neq A_t, H_{t+1}(a) = H_t(a) - \alpha(R_t - \bar{R}_t) \Pr(a)$$

$$H_{t+1}(A_t) \doteq H_t(A_t) + \alpha(R_t - \bar{R}_t)(1 - \pi_t(A_t)), \quad \text{and}$$

$$H_{t+1}(a) \doteq H_t(a) - \alpha(R_t - \bar{R}_t)\pi_t(a), \quad \text{for all } a \neq A_t$$

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What did we learn?

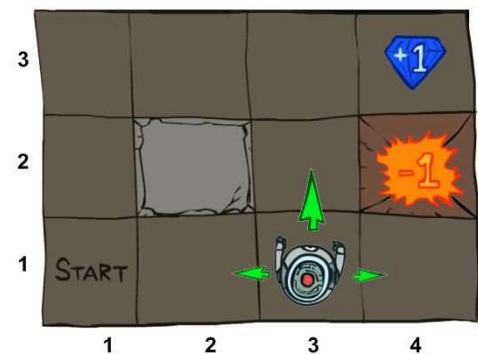
- **Problem:** choose the action that results in highest expected reward
- **Assumptions:** 1. actions' expected reward is unknown, 2. we are confronted with the same problem over and over, 3. we are able to observe an action's outcome once chosen
- **Approach:** learn the actions' expected reward through exploration (value based) or learn a policy directly (policy based), exploit learnt knowledge to choose best action
- **Methods:** 1. greedy + initializing estimates optimistically, 2. epsilon-greedy, 3. Upper-Confidence-Bounds, 4. gradient ascend + soft-max

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A different scenario

- Associative vs. Non-associative tasks ?
- **Policy**: A mapping from situations to the actions that are best in those situations
- (discuss) How do we extend the solution for non-associative task to an associative task?
 - **Approach**: Extend the solutions to non-stationary task to non-associative tasks
 - Works, if the true action values changes slowly
 - What if the context switching between the situations are made explicit?
 - How?
 - Need Special approaches !!!



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Required Readings

1. Chapter-2 of Introduction to Reinforcement Learning, 2nd Ed., Sutton & Barto
2. [A Survey on Practical Applications of Multi-Armed and Contextual Bandits](https://arxiv.org/pdf/1904.10040.pdf), Djallel Bouneffouf, Irina Rish [<https://arxiv.org/pdf/1904.10040.pdf>]

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Thank you !